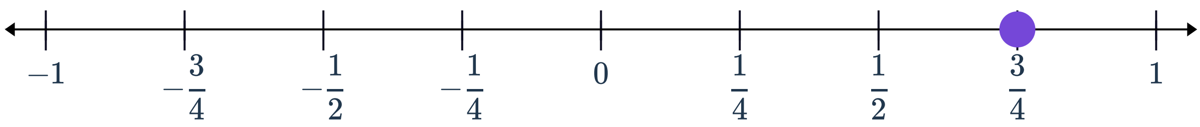
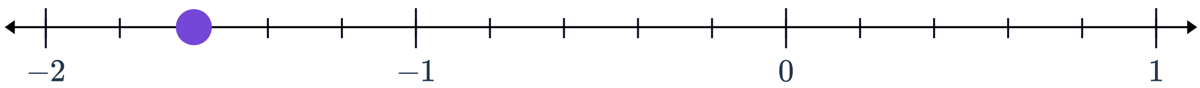
Fractions and the number line

1 State the position of the point plotted on the following number lines as a fraction:

a



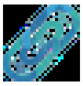
b

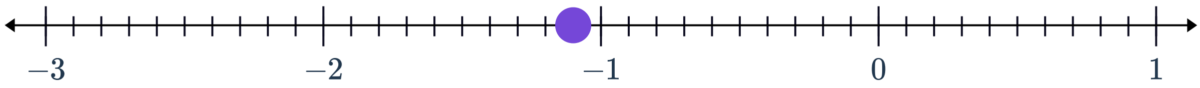


c

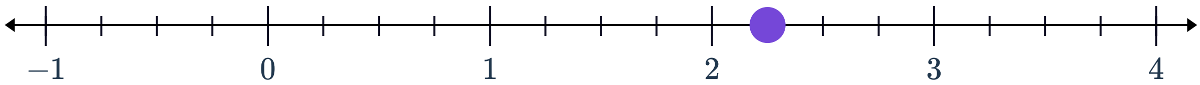
d

e





f

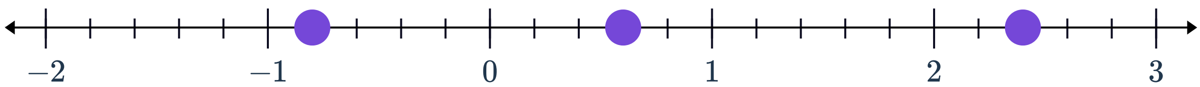


2 For each pair of fractions, state which one is closest to zero:

a $\frac{2}{7}$ and $-\frac{1}{7}$ **b** $-\frac{16}{13}$ and $\frac{7}{13}$ **c** $\frac{2}{11}$ and $-\frac{4}{11}$ **d** $-\frac{1}{4}$ and $\frac{3}{4}$

3 State the greatest number plotted on the following number lines:

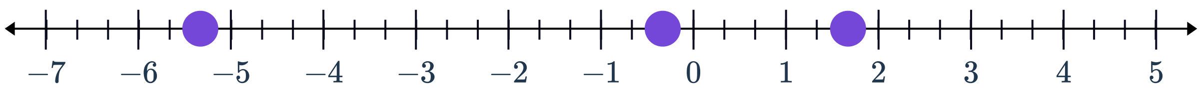
a



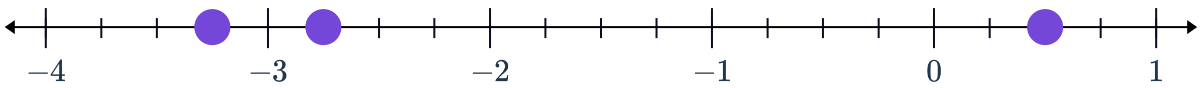
b

4 State the smallest number plotted on the [redacted]ing number lines:

a



b



Fractions with different denominators

5 State the largest number in each set of fractions:

a $-\frac{5}{3}, -\frac{5}{2}, -\frac{7}{3}$

b $-\frac{1}{2}, \frac{2}{3}, \frac{5}{8}$

c $-\frac{3}{8}, -\frac{9}{16}, -\frac{11}{24}$

d $\frac{27}{10}, \frac{13}{5}, \frac{7}{2}$

6 State the smallest number in each set of fractions:

a $-\frac{1}{5}, -\frac{7}{5}, \frac{8}{3}$

b $-\frac{6}{5}, \frac{21}{10}, -\frac{12}{5}$

c $\frac{9}{4}, \frac{22}{7}, \frac{7}{2}$

d $-\frac{9}{5}, -\frac{15}{8}, -\frac{5}{3}$

7 Arrange the following fractions in ascending order: $-\frac{8}{16}, -\frac{2}{16}, -\frac{3}{16}$

8 Arrange the following fractions in descending order: $-\frac{3}{14}, -\frac{19}{14}, -\frac{6}{14}$

9 For each of the following sets of fractions:

i Rewrite all three fractions with the lowest common denominator.

ii Hence, arrange yourthe fractions in ascending order.

a $-\frac{1}{3}, -\frac{5}{8}, -\frac{7}{24}$

b $-\frac{2}{4}, -\frac{6}{10}, -\frac{7}{40}$

c $-1\frac{5}{7}, -\frac{39}{14}, -\frac{19}{7}$

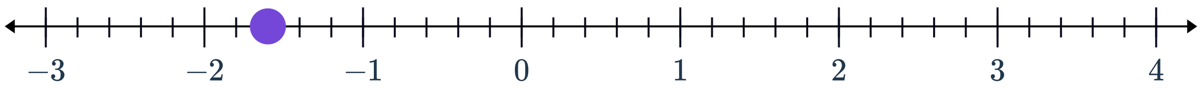
d $-\frac{2}{5}, -\frac{6}{9}, -\frac{11}{45}$

Arithmetic with directed fractions

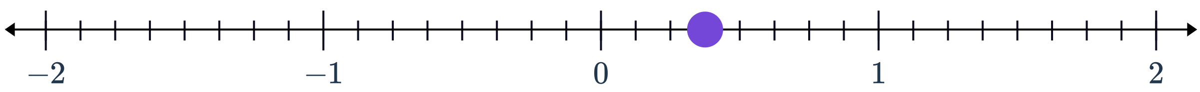
10 Use the plotted point on the following number lines to perform the indicated operation:

a $\frac{3}{5} + \frac{19}{5}$

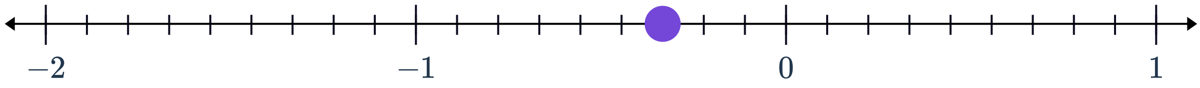
b $-\frac{8}{5} + \frac{19}{5}$



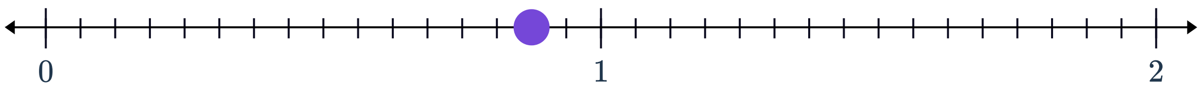
c $\frac{3}{8} - \frac{5}{8}$



d $\frac{5}{9} - \frac{8}{9}$



e $\frac{14}{16} - \frac{3}{16}$



11 Evaluate:

a $-\frac{7}{3} + \frac{17}{3}$

b $-\frac{26}{3} + \frac{16}{3}$

c $\frac{3}{5} - \frac{14}{5}$

d $\frac{5}{11} + \left(-\frac{9}{11}\right)$

e $-\frac{5}{9} + \left(-\frac{11}{9}\right)$

f $2\frac{2}{9} - \left(-\frac{5}{9}\right)$

g $-\frac{2}{9} - \left(-\frac{3}{5}\right)$

h $-\frac{2}{3} + \left(-\frac{1}{9}\right)$

Applications

12 If Patricia travels $4\frac{2}{5}$ km west, then $6\frac{3}{5}$ km east, find how far she is from her starting point.

13 This month, John spent $\frac{1}{4}$ of his allowance on food, $\frac{1}{8}$ on school projects, $\frac{3}{16}$ on transportation. What fraction of his allowance is left?

14 The average water level at a port is marked at 0 m, and a tide scale is used to measure the water level as it moves up and down with the tide.

When Jack begins his shift at 11 am, the tide scale reads $\frac{2}{5}$ m. Over the next 7 hours, the tide falls by $\frac{2}{3}$ m, and his shift ends. State the reading of the tide scale at the end of his shift.

15 The largest pumpkin in Shawn's farm weig[illegible] $\frac{3}{8}$ kg. This is $1\frac{5}{6}$ kg more than the average weight of pumpkins in his farm. Find the average weight of the pumpkins from Shawn's farm.

16 Willy and Bill are selling lemonade in the school fair. They have prepared $24\frac{3}{4}$ cups, which costed them $\$18$. Each serving of lemonade contains $1\frac{1}{4}$ cups, which they sell at $\$2$.

a At the end of day, Willy and Bill checked the jug and there is still $3\frac{1}{2}$ cups left over. How many cups of lemonade did they sell that day?

b Did Willy and Bill make a profit? Explain your answer.

17 Diane needs to purchase flour for a cooking class in school. The class is going to prepare a chocolate cake which requires $2\frac{1}{2}$ kg of flour, battered fish which requires $4\frac{3}{4}$ kgof flour, and banana bread which requires $3\frac{1}{4}$ kg of flour.

If the school already had 5 kg of flour, how many cups of flour Diane must buy?